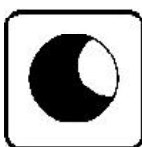
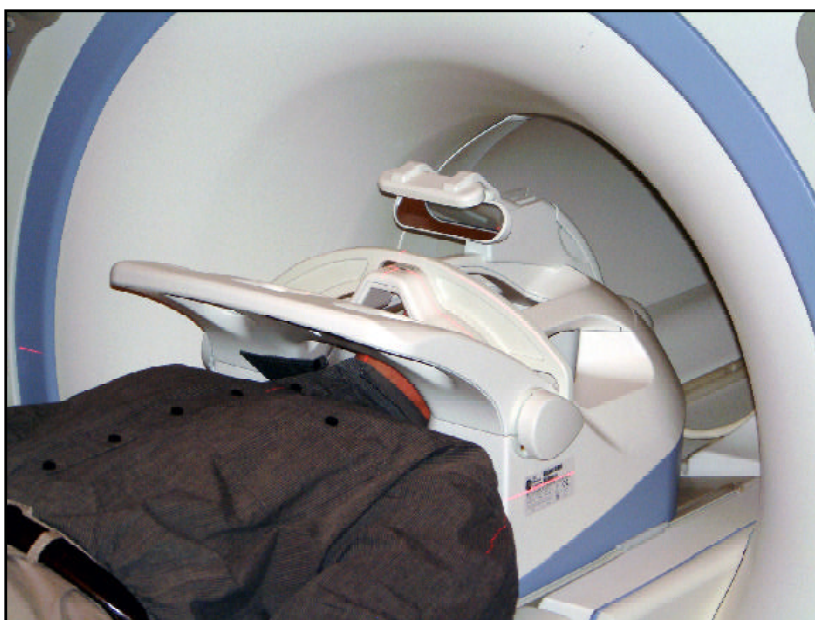


Operator's Manual

8 Channel Neurovascular Array for the GE 1.5T Signa Infinity with Excite Technology



Manufactured by:

MRI Devices Corporation
1515 Paramount Drive, Suite A
Waukesha, Wisconsin 53186 USA



Distributed by:

GE Medical Systems



First Edition
October 2002
©2002 MRI Devices Corporation

All rights reserved. No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language in any form by any means without the written permission of MRI Devices Corporation.

Licenses and Trademarks

The MRI Devices Logo is a registered trademark of MRI Devices Corporation.

"Signa Infinity with Excite Technology" is a registered trademark of the General Electric Company.

Proper performance of this coil is guaranteed only while the coil is being used on the MR system (hardware/software level) specified at the time of purchase. Upgrades or other modifications to the system software and/or hardware may affect compatibility. Prior to upgrading your MR system, please contact your GE Medical Systems representative to discuss coil compatibility issues. Failure to do so may void your warranty.

Medical Device Directive

These products conform with the requirements of council directive 93/42/EEC concerning medical devices, when they bear the following CE Mark of Conformity:



European Representative

GE Medical Systems Europe
Quality Assurance Manager
BP 34
F 78533 BUC CEDEX
France
Telephone: +33 1 30 70 40 40



Attention, Consult Accompanying Documents



Type BF Equipment



Class II Equipment

NOTICE:

TRANSPORT AND STORE THIS PRODUCT UNDER THE FOLLOWING ENVIRONMENTAL CONDITIONS ONLY, FOR A PERIOD NOT EXCEEDING 4 WEEKS:

AMBIENT TEMPERATURE OF -40°C to +60°C
RELATIVE HUMIDITY OF 10% TO 100% (Non-Condensing)

ATMOSPHERIC PRESSURE ON 765 hPa TO 1011 hPa

Caution:
Federal law restricts this device to sale, distribution, and use by or on the order of a physician.

Introduction

This manual describes the safety precautions, features, use and care of the MRI Devices 8 Channel Neurovascular Array, compatible with the GE 1.5T Signa Infinity with Excite Technology. Please review this manual thoroughly before using the device.

If you have any questions or comments on this manual, or need any assistance with the use of the product, please contact your GE Medical Systems representative at:

1-800-GE CARES

Explanation of Symbols

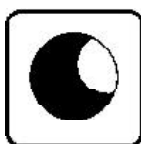
 Medical Device Directive - These products conform with the requirements of council directive 93/42/EEC concerning medical devices, when they bear the CE Mark of Conformity.

 Attention, Consult Accompanying Documents.

 Type BF Equipment

 Class II Equipment

 Do not cross or loop cables. Arcing and patient burns could result.



Manufactured by:

MRI Devices Corporation
1515 Paramount Drive, Suite A
Waukesha, Wisconsin 53186 USA

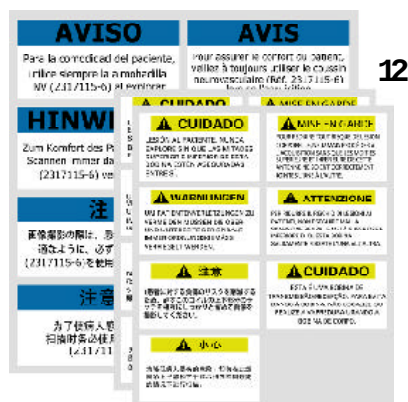
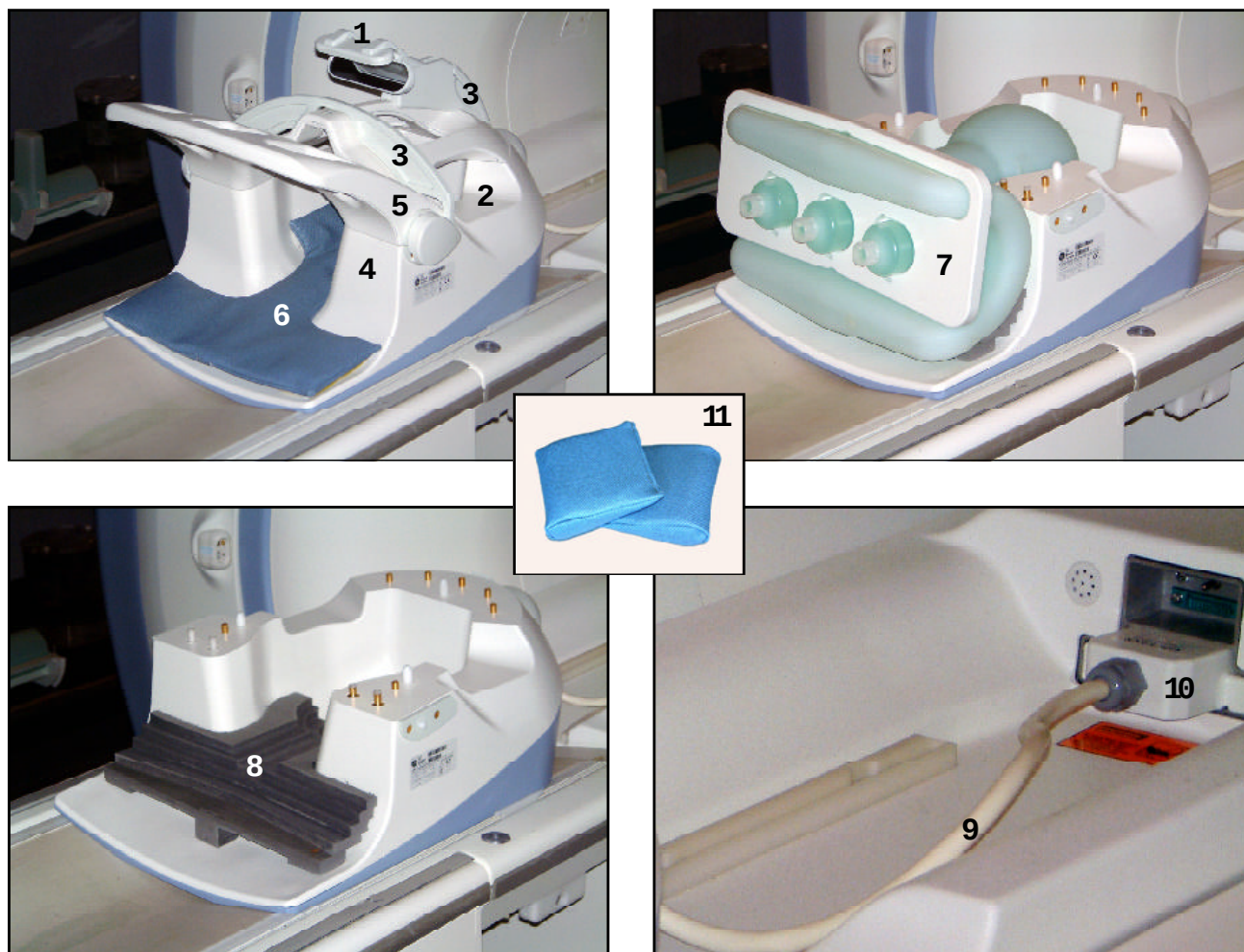
Distributed by:

GE Medical Systems



8 Channel Neurovascular Array

The 8 Channel Neurovascular Array consists of the following parts. Please inspect upon receipt to make sure all parts have arrived and are in good order. Use this guide to refer to part names throughout this manual.



- | | | | |
|---|---------------------|----|--------------------------|
| 1 | Mirror | 7 | Phantom Kit |
| 2 | Window | 8 | Phantom Positioner* |
| 3 | Fastener | 9 | Coil Cable |
| 4 | Posterior Coil | 10 | Coil Connector |
| 5 | Anterior Coil | 11 | Wedge Pads |
| 6 | Patient Comfort Pad | 12 | Translated Safety Labels |

*Phantom Positioner is part of Phantom Kit

Table of Contents

Introduction	3
Explanation of Symbols	3
8 Channel Neurovascular Array	4
Chapter 1 - Safety	6
Training	6
Quality Assurance	6
Indications	6
Contra-indications	6
Precautions	6
Cautions	7
Emergency Procedures	7
Chapter 2 - Coil Configuration Quickstep Guide	8
8NVHEAD_A	8
8NVHEAD ONE RCVR	8
8NVHEAD NECK_A	8
8NVHEAD NECK	8
8NVARRAY_A	9
8NVANGIO_A	9
8NVANGIO LAT	9
8NVANGIO MED	9
Chapter 3 - Using the 8 Channel Neurovascular Array	10
Positioning the Neurovascular Array on the Patient Cradle	10
Positioning the Patient	11
Adjusting the Patient Mirror	12
Connecting the Cable	12
Establishing the Patient LANDMARK	13
Chapter 4 - Scanning	14
Selecting the Coil	14
Auto Pre-scan	14
Localizing with the Body Coil	14
IMPORTANT! Using Autoslim	14
Magnetization Transfer Techniques	14
Protocols	14
Chapter 5 - Quality Assurance	15
SNR Protocol	15
Chapter 6 - Installation and Maintenance	30
Installation and Configuration	30
Translated Safety Labels	30
Cleaning	30
Storage	30
Replaceable Accessories	31

Chapter 1 - Safety

Training

This manual contains detailed information on the set-up, positioning and use of your MRI Devices Corporation coil. The instructions should be read carefully and thoroughly before attempting to scan patients with the coil.

Quality Assurance

The procedure described in the Quality Assurance Section of this manual should be performed upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals.

Indications

The coil is indicated for use, on the order of a physician, in conjunction with an MR scanner as an accessory to produce general purpose images.

Contra-indications

The operator should be aware of the following contraindications for use related to the strong magnetic field of the MR system:

 Scanning is contraindicated for patients who have electrically, magnetically or mechanically activated implants (for example, cardiac pacemakers). The magnetic and electromagnetic fields produced by the MR System and coil may interfere with the operations of these devices.

 Scanning patients with intracranial aneurysm clips is contraindicated.

Precautions

Precautions should be taken when scanning patients with the following conditions:

 Greater than normal potential for cardiac arrest

 An increased likelihood for developing seizures or claustrophobia

 Unconscious, heavily sedated, or confused patients

 Patients with whom reliable communications cannot be maintained

Cautions

The following general warning statements apply to scanning with a magnetic resonance system. For further details, review the warnings in your MR system Operators Manual.



Do not cross or loop cables. Arcing and patient burns could result. Route cables out of the magnet so that they do not touch the patient.



If a patient complains of burning, tingling, stinging, or other similar 'burn'-type sensations, promptly stop the scan procedure, examine the patient, and contact the responsible physician before continuing the procedure. Pay special attention to very young, sedated, or other compromised patients who may not be able to communicate effectively.



Patients with ferromagnetic metal should not be scanned because the magnetic field may interact with implanted surgical clips or other ferromagnetic materials.



Persons with cardiac pacemakers or other implanted electronic devices should not enter the magnetic field zone delineated by the MR system manufacturer.



There is a risk to scanning feverish or decompensated cardiac patients.



Facial makeup should be removed before scanning because it may contain metal flakes which can cause skin and eye irritation. Permanent eyeliner tattoos may cause eye irritation due to ferromagnetic particles.



Patients who work in environments in which there is a risk of having embedded metallic fragments in or near the eye should be carefully screened before undergoing an MR exam.



Visually inspect the white insulator jacket and the coil connector with each use. If the insulator is broken, or the cable is frayed at the connector, immediately discontinue use of the device.



Emergency Procedures

In the unlikely event that a coil creates smoke, sparks or makes an unusually loud noise, or if the patient requires emergency assistance:

- Stop the scan if one is in progress.
- Remove the patient from the scan room if medical treatment is needed.

Chapter 2 - Coil Configuration Quickstep Guide

8NVHEAD_A

- ASSET Compatible
- Highest SNR for brain imaging
- Maximum FOV 24CM
- SCIC Recommended
- Not Compatible with spectroscopy

Applications: Brain / Cerebellum Imaging

8NVHEAD ONE RCVR

- **NOT** ASSET Compatible
- Highest uniformity for brain imaging
(decreased SNR compared to 8NVHEAD_A)
- Maximum FOV 24CM
- SCIC Recommended

Applications: Brain / Cerebellum Imaging
Spectroscopy

8NVHEAD NECK_A

- ASSET Compatible
- Increased S/I coverage as compared to 8NVHEAD_A
- Maximum FOV 28CM
- SCIC Recommended

Applications: Brain Imaging
(when extended S/I coverage is needed)
Soft-tissue neck Imaging

8NVHEAD NECK

- **NOT** ASSET Compatible
- Increased S/I coverage as compared to 8NVHEAD_A
- More uniformity, but decreased SNR as compared to 8NVHEADNECK_A
- Maximum FOV 28CM
- SCIC Recommended

Applications: Brain Imaging
(when extended S/I coverage is needed)
Soft-tissue neck Imaging

8NVARRAY_A

- ASSET Compatible
- Maximum FOV 40CM
- SCIC Recommended
- Not recommended for angiographic exams

Applications: Cervical Spine
Soft tissue Neck

8NVANGIO_A

- ASSET Compatible
- Optional Configuration. For install, contact your GE Field Service Representative
- Maximum FOV 40CM
- SCIC Recommended
- Improved uniformity as compared to 8NVARRAY_A

Applications: Carotid angiographic studies with ASSET

8NVANGIO LAT

- **NOT** ASSET Compatible
- Largest FOV coverage for Angio imaging
- Maximum FOV 40CM
- SCIC Recommended

Applications: Angio imaging of the Brain and Neck
Brachial Plexus

8NVANGIO MED

- **NOT** ASSET Compatible
- Reduced **R>L** coil coverage to reduce phase wrap from shoulder anatomy
- Maximum FOV 40CM
- SCIC Recommended

Applications: All Angio imaging of the head and neck
Soft-tissue neck Imaging

Chapter 3 - Using the 8 Channel Neurovascular Array

Positioning the Neurovascular Array on the Patient Cradle

The 8 Channel Neurovascular Array is designed for imaging of the head, neck and chest.

The array must be placed on the patient cradle with the cable toward the magnet. The 8 Channel Neurovascular Array is designed for head first exams only!

The Neurovascular Array base is designed to rest directly on the patient table. To align the coil in a fixed position, insert the tabs on the bottom of the coil into the slots on the patient table.



The patient comfort pad should be centered in the coil. The pad serves to provide additional patient comfort during the scan, helps center the patient in the coil, and is recommended for optimal coil performance.

Positioning the Patient

Position the patient in the coil such that the contour of the neck fits the contoured shape of the coil. The patient's shoulders should rest comfortably just outside the coil.



Carefully aligning the contacts, put the anterior portion of the coil in place.



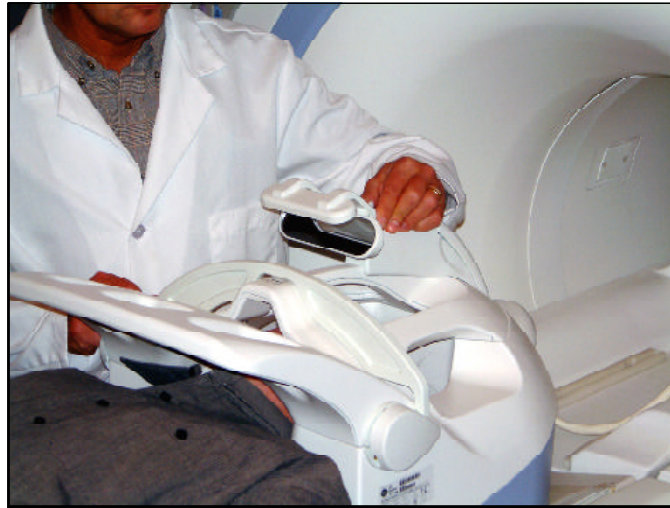
If necessary, insert wedge pads to help immobilize the patient's head.



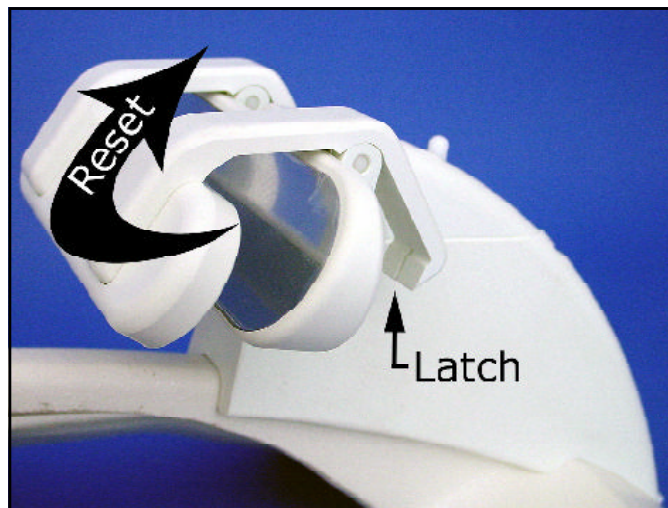
Secure the anterior section of the coil by fastening both latch levers.

Adjusting the Patient Mirror

The mirror is positioned by sliding in the superior or inferior direction until the patient has a clear view of the control room.



If the mirror comes into contact with the bore, a special mirror design allows the upper mirror to "break away," as shown below. Should this happen, reset the upper mirror by pushing it up until the latches are secured.



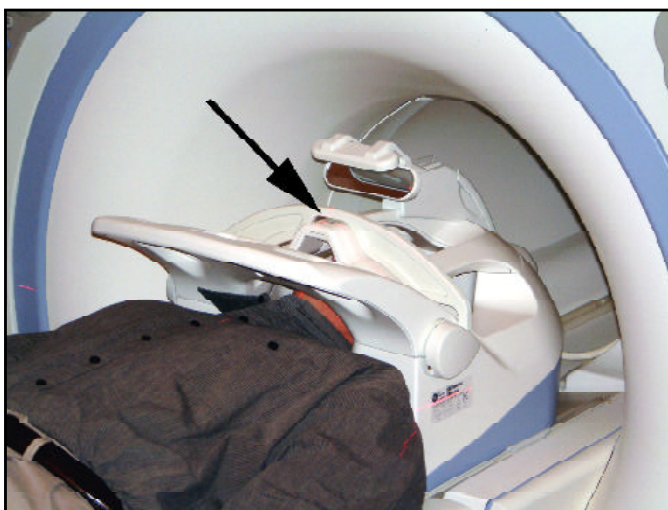
Connecting the Cable



Connect the coil to the system using the special phased array 8 channel coil connector.

Establishing the Patient LANDMARK

Use the windows on the coil housing to ensure the patient is centered properly. Landmark on the coil cross hairs and advance the patient into the magnet for scanning.



Chapter 4 - Scanning

Selecting the Coil

There are eight coil configurations available for use with the 8 Channel Neurovascular Array. For more information see Chapter 2 - Coil Configuration Quickstep Guide.

Auto Pre-scan

Generally, the Signa Infinity with Excite Technology System auto pre-scan will function normally when used with the 8 Channel Neurovascular Array. If auto pre-scan returns the message "auto pre-scan failed to complete normally," then manual pre-scan may be required to acquire the image. Consult your GE Signa Infinity with Excite Technology operator's manual for manual pre-scan procedures.

Localizing with the Body Coil

The body coil may be used at any time while the 8 Channel Neurovascular Array is connected to the scanner. This allows a large FOV (40 cm.) body coil localizer to be performed, if desired.

IMPORTANT! Using Autoshim

AUTOSHIM is a feature of the software AUTOPRESCAN to improve image quality. It does this by improving the magnetic field homogeneity within the FOV selected. The improvement in image quality is often dramatic when the selected FOV is far off center, and when acquiring FSE or T2 weighted scans, or FAT SAT images.

Magnetization Transfer Techniques

Magnetization transfer techniques are available with the GE Signa Angio software.

Protocols

Refer to the GE protocols on your system.

Chapter 5 - Quality Assurance

SNR Protocol

This section contains the Coil Imaging Performance Verification procedure from the GE 1.5T 8 Channel Neurovascular Array Service Manual. The information is duplicated complete with Service Manual section subheadings and numbering.

3-2 Coil Imaging Performance Verification

3-2-1 Data Collection

1. End the previous exam if this has not already been done.
2. Attach the base of the 8 channel neurovascular array to the cradle in the same position on the cradle as the standard head coil as shown in Figure 3-2-1-1 and remove the top.
3. Note the serial number of the coil.
4. Place phantom positioner on coil as shown in Figure 3-2-1-2.
5. Place phantom on positioner as shown in Figure 3-2-1-3.
6. Attach top to coil as shown in Figure 3-2-1-4.
7. Establish an axial landmark on the line on top of the coil. Advance to scan.
8. On the service desktop open a command window and navigate to the **/usr/local/bin/CoilsTools** directory.
9. Enter the command **run8nvarrayQA** and press the enter key.
10. When scanning is completed and images have been extracted (messages stop scrolling in the command window and the prompt returns approximately 5 minutes) note the exam number and continue with the SNR Analysis section.

Note: Examples of images acquired during execution of the tool are provided at the end of this procedure in section 3-2-4 Example Images.

Series 1: Body localizer for positioning ROIs on head section during 8NVHEAD acquisition. 1 image. See Figure 3-2-4-1.

Series 2: 8NVHEAD mode scan, 1 slice, 2 scans of 9 images each. See Figure 3-2-4-2.

Series 3: Body localizer for positioning ROIs on chest (slice 1), neck (slice 2) and head (slice 3) sections during 8NVARRAY acquisition. 3 images. See Figure 3-2-4-3.

Series 4: 8NVARRAY mode, 3 slices, 2 scans of 27 images each. See Figures 3-2-4-4, 3-2-4-5, and 3-2-4-6.

Figure 3-2-1-1



Remove Coil Top

Figure 3-2-1-2



Place Phantom Positioner on coil

Figure 3-2-1-3



Place Phantom on Positioner

Figure 3-2-1-4



Place Top on Coil

Figure 3-2-1-5



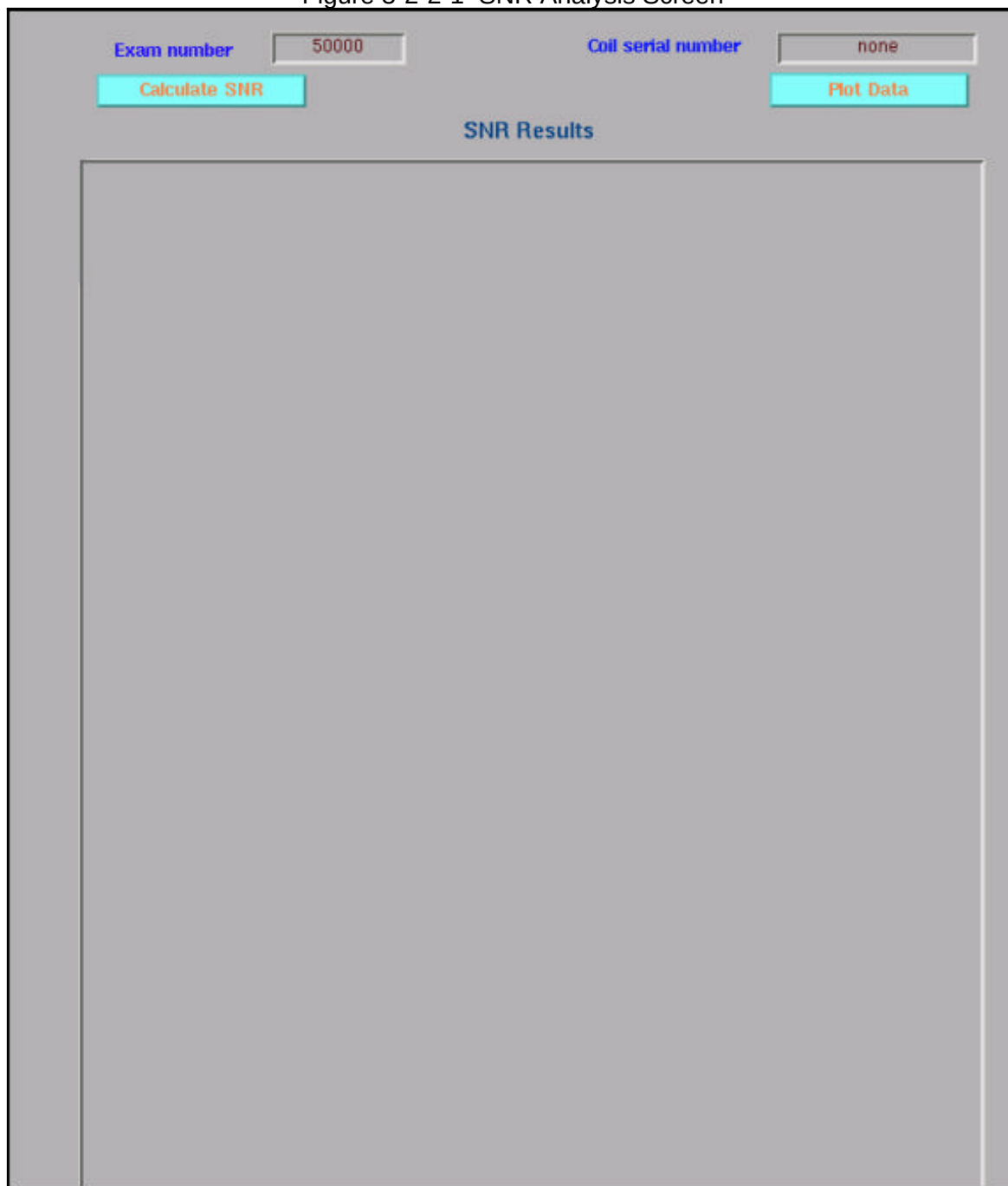
Correct Bottle Placement

NOTE: For correct bottle placement, insert the bottom of each phantom bottle fully, until it touches the curved inside surface of the main phantom. Refer to Figure 3-2-1-5.

3-2-2 SNR Analysis

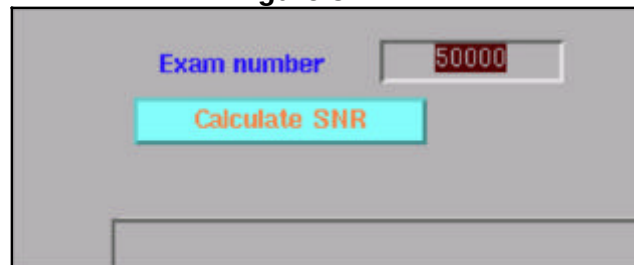
1. In the command window enter **snr_for_8nvarray_QA** and press enter. A screen like that shown in Figure 3-2-2-1 should appear.

Figure 3-2-2-1 SNR Analysis Screen



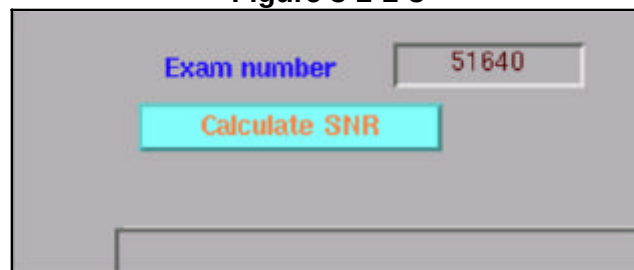
The image shows a software interface for SNR analysis. At the top, there are two input fields: "Exam number" with the value "50000" and "Coil serial number" with the value "none". Below these are two buttons: "Calculate SNR" and "Plot Data". In the center, the text "SNR Results" is displayed above a large, empty rectangular area, which is likely intended for a plot or data output.

2. In the Exam number window, enter the exam number as shown in Figures 3-2-2-2 and 3-2-2-3 (The Delete key must be used instead of Backspace).
3. In the Coil serial number window, enter serial number and bay number as shown in Figures 3-2-2-4 and 3-2-2-5 (Optional).

Figure 3-2-2-2

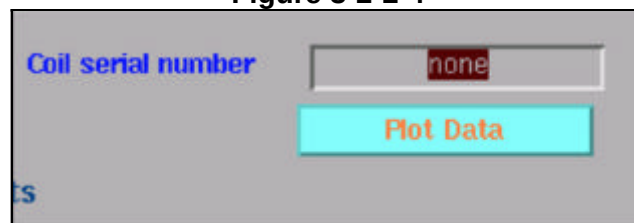
Exam number 50000

Calculate SNR

Highlight Digits for Exam Number**Figure 3-2-2-3**

Exam number 51640

Calculate SNR

Type In Exam Number**Figure 3-2-2-4**

Coil serial number none

Plot Data

Highlight Coil Serial Number**Figure 3-2-2-5**

Coil serial number bay4a U12345

Plot Data

Enter Coil Serial Number and Bay Number

4. Click the Calculate SNR button as shown in Figure 3-2-2-6.

Figure 3-2-2-6

Exam number 51640

Calculate SNR

Click Calculate SNR Button

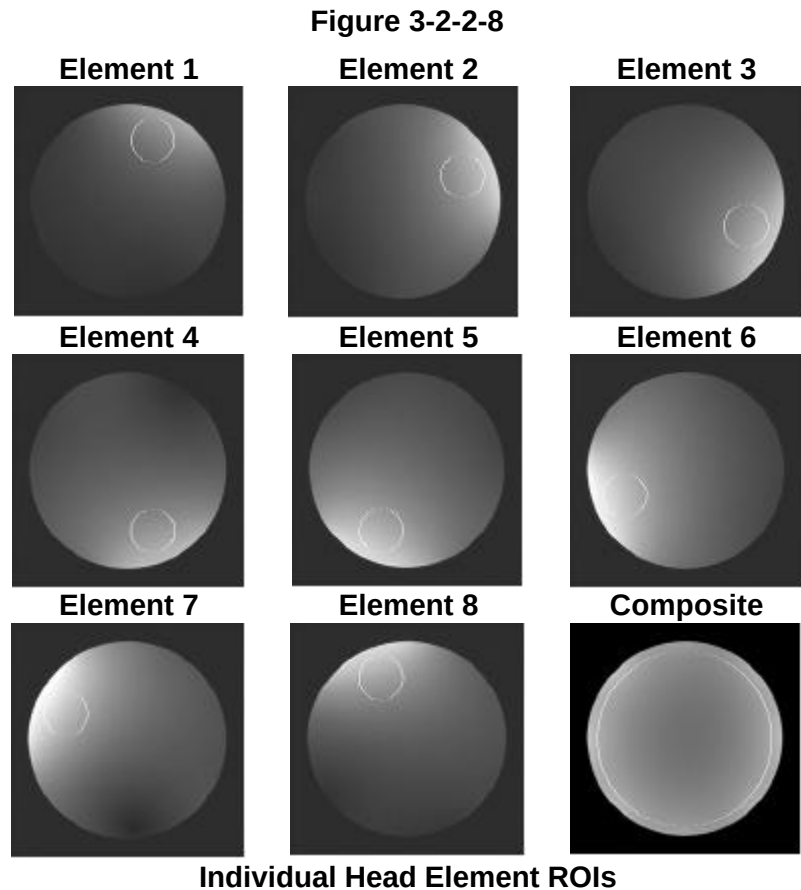
5. Results will appear as shown in Figure 3-2-2-7.

Figure 3-2-2-7

SNR Results			
8 Channel Neurovascular Array Quality Assurance Test			
serial number bay4a U12345			
8NVHEAD mode *****			
phantom center =(127,129)			
Individual Element NEMA SNR from composite images			
ROI	signal	noise	snr
1	758.9	4.73	226.8
2	871.0	4.82	255.5
3	889.4	4.91	256.1
4	867.1	4.88	251.1
5	857.1	5.12	236.8
6	876.3	5.13	241.6
7	900.9	4.87	261.5
8	802.5	4.66	243.7
Composite SNR Using NEMA Method			
ROI	signal	noise	snr
C	825.7	5.03	222.0
8NVARRAY head area *****			
phantom center =(127,129)			
Individual Element NEMA SNR from composite images			
ROI	signal	noise	snr
1	873.9	4.54	272.3
2	965.1	4.35	313.5
3	988.0	4.02	347.9
4	1022.0	4.33	333.5
5	1069.4	4.15	364.3
6	1119.4	4.53	349.8
7	1083.4	4.78	320.2
8	938.8	4.93	269.2
Composite SNR Using NEMA Method			
ROI	signal	noise	snr
C	942.5	4.37	291.4
8NVARRAY neck area *****			
phantom center =(124,129)			
Composite SNR Using NEMA Method			
ROI	signal	noise	snr
C	722.5	5.28	184.9
8NVARRAY chest mode *****			
phantom center =(122,134)			
Individual Element NEMA SNR from composite images			
ROI	signal	noise	snr
1	237.0	4.70	71.3
2	211.0	4.35	68.7
3	283.6	4.84	82.9

SNR Results

6. Table 3-2-3-1 shows the test results that correspond to each test.



Note: The results are also stored in a file in the /usr/g/service/data directory. The filename will be similar to 8nvarray_51897_c7ga0522.coilqa where 51897 is the exam number.

7. Record the coil serial number and the associated filename in Table 3-2-3-3 for future reference.

Note: The analysis tool can be used to process multiple runs by entering a new exam number and coil serial number and pressing the Calculate SNR button again. It is not necessary to exit the analysis tool after each run.

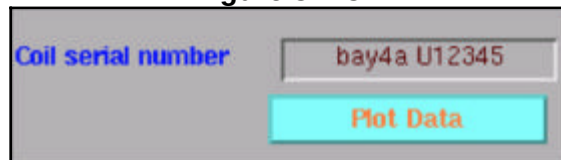
8. Proceed to the next section for instructions to plot the data.

3-2-3 Plot Results

Note: Perform steps 1 through 3 only if the SNR tool is not already running.

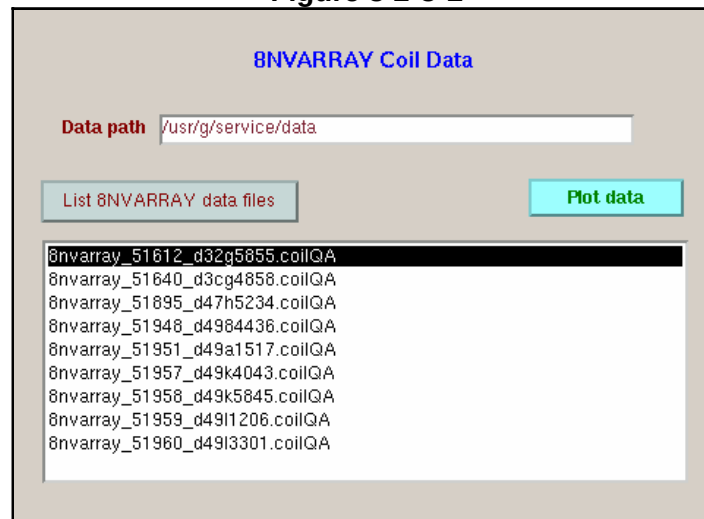
Note: Table 3-2-3-1 explains the output of the tool. Table 3-2-3-2 provides SNR acceptance limits. Table 3-2-3-3 provides a data sheet for recording file names for future reference.

1. Open a command window on the tools desktop.
2. Navigate to the /usr/local/bin/CoilsTools directory.
3. In the command window enter **snr_for_8nvarray_QA** and press enter. A screen like that shown in Figure 3-2-2-1 should appear.
4. Click on the plot data button as shown in Figure 3-2-3-1.
5. Click on the “List 8NVARRAY data files” button as shown in Figure 3-2-3-2.
6. Click on the name of the file to be plotted as shown in Figure 3-2-3-3.
7. Click on the plot data button as shown in Figure 3-2-3-4 to process the file.
8. Example of a nominal coil is shown in Figure 3-2-3-5.

Figure 3-2-3-1

Coil serial number

Plot Data

Start Plot Tool**Figure 3-2-3-2**

8NVARRAY Coil Data

Data path

List 8NVARRAY data files **Plot data**

8nvarray_51612_d32g5655.coilQA
8nvarray_51640_d3cg4858.coilQA
8nvarray_51895_d47h5234.coilQA
8nvarray_51948_d4984436.coilQA
8nvarray_51951_d49a1517.coilQA
8nvarray_51957_d49k4043.coilQA
8nvarray_51958_d49k5845.coilQA
8nvarray_51959_d49i1206.coilQA
8nvarray_51960_d49i3301.coilQA

List Files

Figure 3-2-3-3

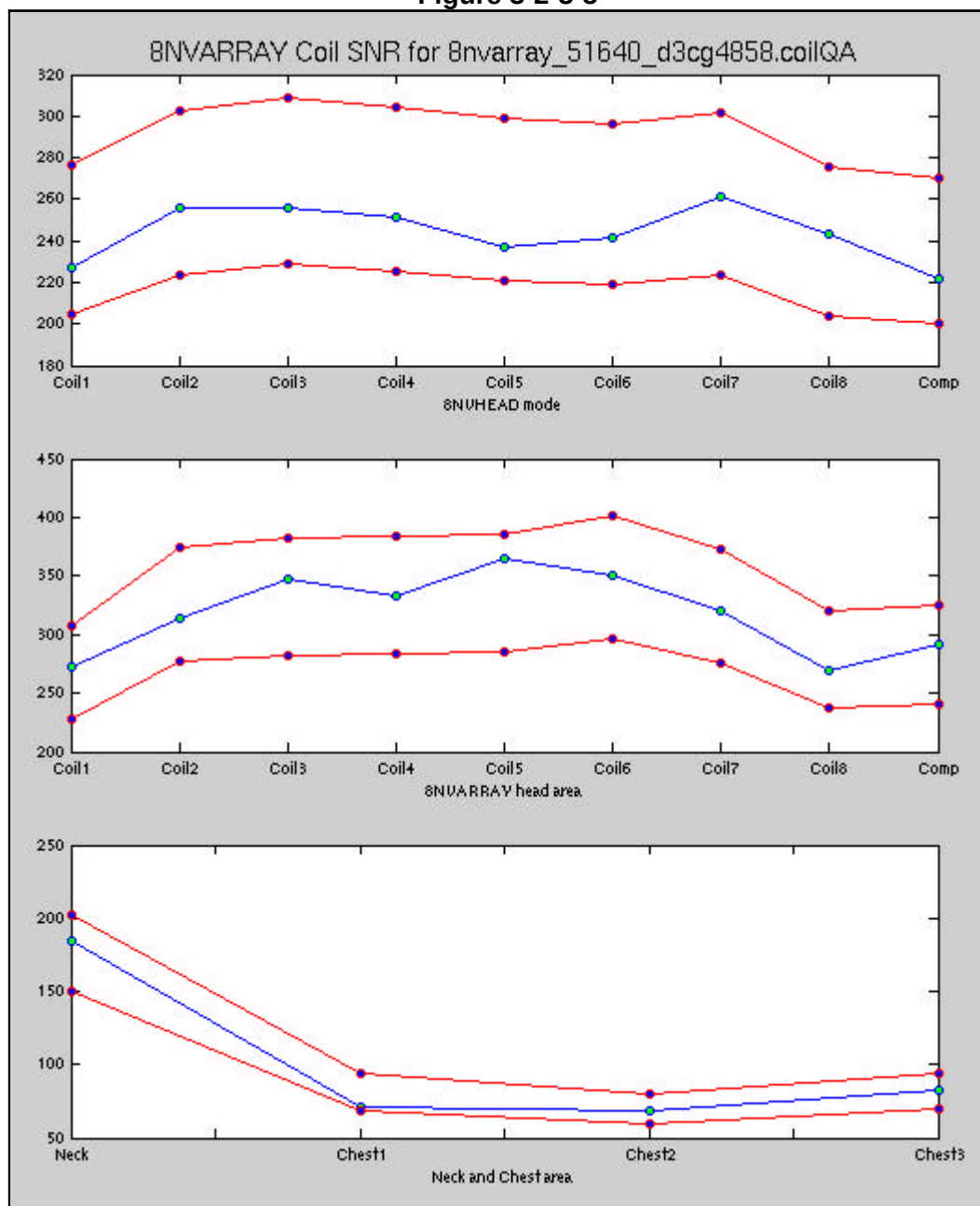
BNVARRAY Coil Data

Data path

8nvarray_51612_d32g5855.coilQA
8nvarray_51640_d3cg4858.coilQA
8nvarray_51895_d47h5234.coilQA
8nvarray_51948_d4984436.coilQA
8nvarray_51951_d49a1517.coilQA
8nvarray_51957_d49k4043.coilQA
8nvarray_51958_d49k5845.coilQA
8nvarray_51959_d49l1206.coilQA
8nvarray_51960_d49l3301.coilQA

Select File**Figure 3-2-3-4****Process File**

Figure 3-2-3-5



Results Compared to Upper and Lower Limits

Note: Red lines show upper and lower acceptance limits. Blue line shows test results.

TABLE 3-2-3-1 EXAMPLE OUTPUT OF TOOL

8 Channel Neurovascular Array Quality Assurance Test			
serial number bay4a U12345			
8NVHEAD mode *****			
phantom center =(127,129)			
Individual Element NEMA SNR from composite images			
ROI	signal	noise	snr
1	758.9	4.73	226.8
2	871.0	4.82	255.5
3	889.4	4.91	256.1
4	867.1	4.88	251.1
5	857.1	5.12	236.8
6	876.3	5.13	241.6
7	900.9	4.87	261.5
8	802.5	4.66	243.7
Composite SNR Using NEMA Method			
ROI	signal	noise	snr
C	825.7	5.03	222.0
8NVARRAY head area *****			
phantom center =(127,129)			
Individual Element NEMA SNR from composite images			
ROI	signal	noise	snr
1	873.9	4.54	272.3
2	965.1	4.35	313.5
3	988.0	4.02	347.9
4	1022.0	4.33	333.5
5	1069.4	4.15	364.3
6	1119.4	4.53	349.8
7	1083.4	4.78	320.2
8	938.8	4.93	269.2
Composite SNR Using NEMA Method			
ROI	signal	noise	snr
C	942.5	4.37	291.4
8NVARRAY neck area *****			
phantom center =(124,129)			
Composite SNR Using NEMA Method			
ROI	signal	noise	snr
C	722.5	5.28	184.9
8NVARRAY chest mode *****			
phantom center =(122,134)			
Individual Element NEMA SNR from composite images			
ROI	signal	noise	snr
1	237.0	4.70	71.3
2	211.0	4.35	68.7
3	283.6	4.84	82.9

Using ROIs shown in Figure 3-2-2-8 signal is measured in each of the two composite images of the head area and averaged. The same ROI for each element is used to measure noise in the filtered difference image.

Composite image results

Using ROIs shown in Figure 3-2-2-8 signal is measured in each of the two composite images of the head area and averaged. The same ROI for each element is used to measure noise in the filtered difference image.

Composite image results

Composite image results

Composite image results measured on a circular ROI (each of 3 bottles in chest are

TABLE 3-2-3-2 SNR LIMITS

			nominal	USL	LSL
8NVHEAD mode	head	elem1	240.6	277	205
		elem2	262.9	302	223
		elem3	268.8	309	228
		elem4	264.8	304	225
		elem5	259.6	299	221
		elem6	257.3	296	219
		elem7	262.5	302	223
		elem8	239.8	276	204
		comp	231.4	266	197
8NVARRAY mode	head	elem1	267.5	308	227
		elem2	325.4	374	277
		elem3	332.3	382	282
		elem4	334.0	384	284
		elem5	335.1	385	285
		elem6	348.6	401	296
		elem7	324.6	373	276
		elem8	278.8	321	237
		comp	285.0	328	242
		neck	175.8	202	149
		chest 1	79.6	92	68
		chest 2	69.4	80	59
		chest 3	80.6	93	68

TABLE 3-2-3-3 DATA SHEET

Coil Serial Number	Filename

3-2-4 Example Images

Figure 3-2-4-1
Receiver 1
8NVHEAD Mode Localizer Image

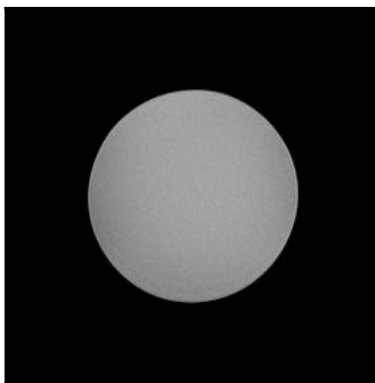


Figure 3-2-4-2 8NVHEAD Mode Signal Images

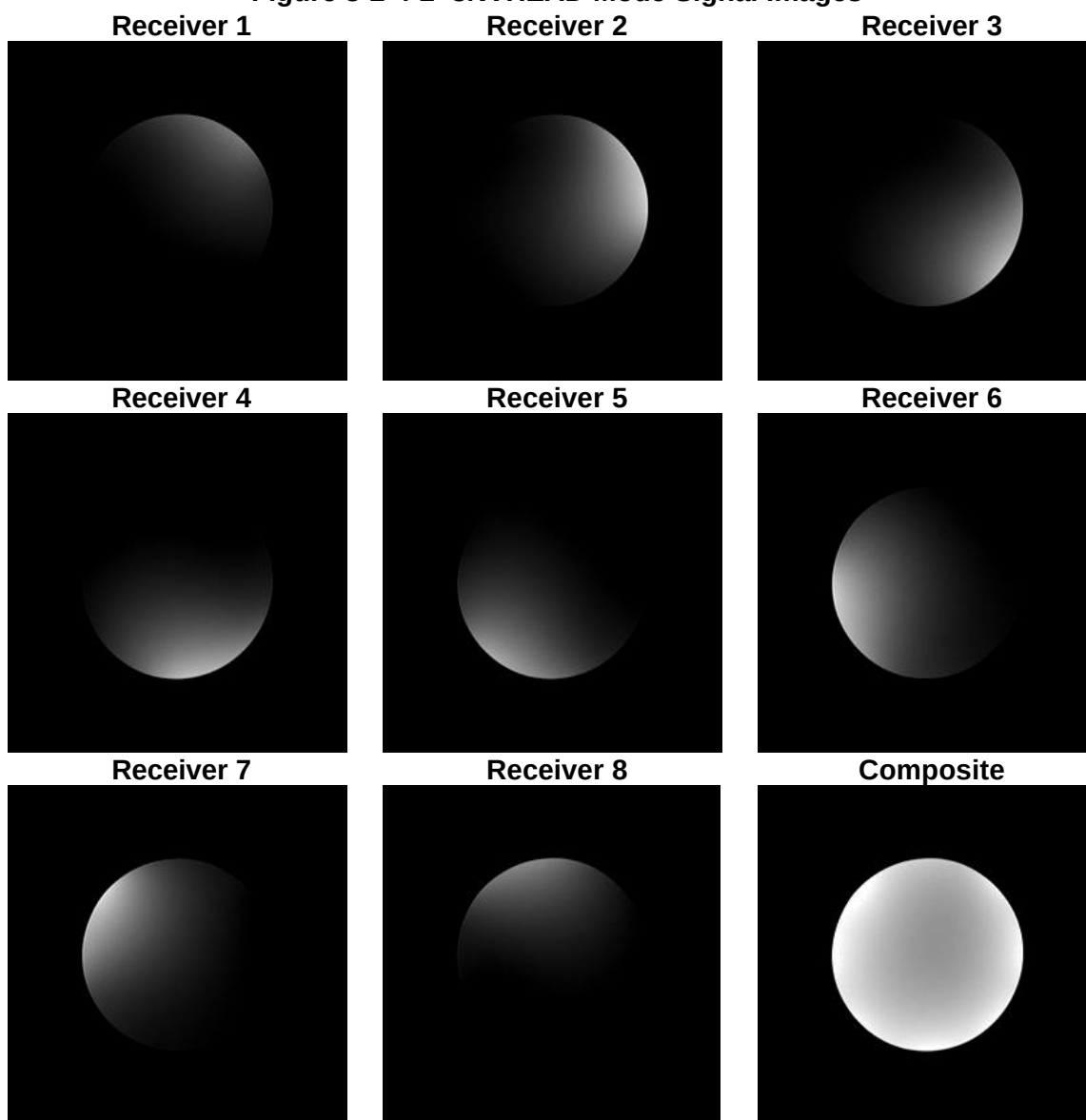
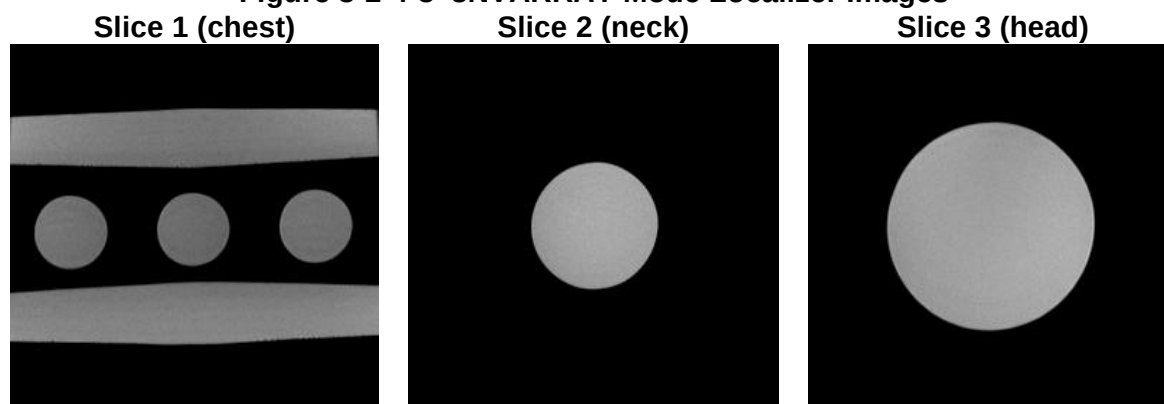
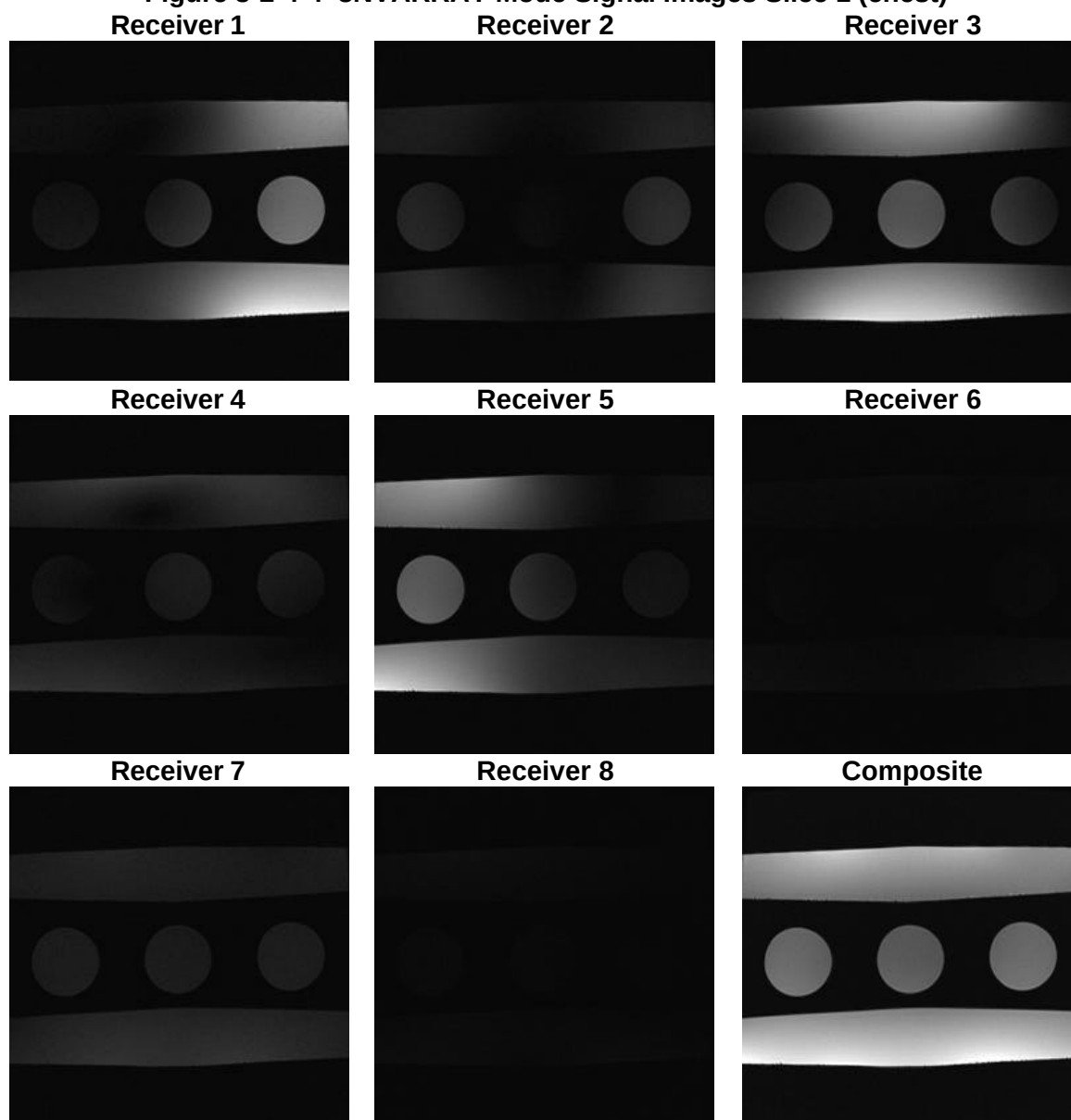


Figure 3-2-4-3 8NVARRAY Mode Localizer Images**Figure 3-2-4-4 8NVARRAY Mode Signal Images Slice 1 (chest)**

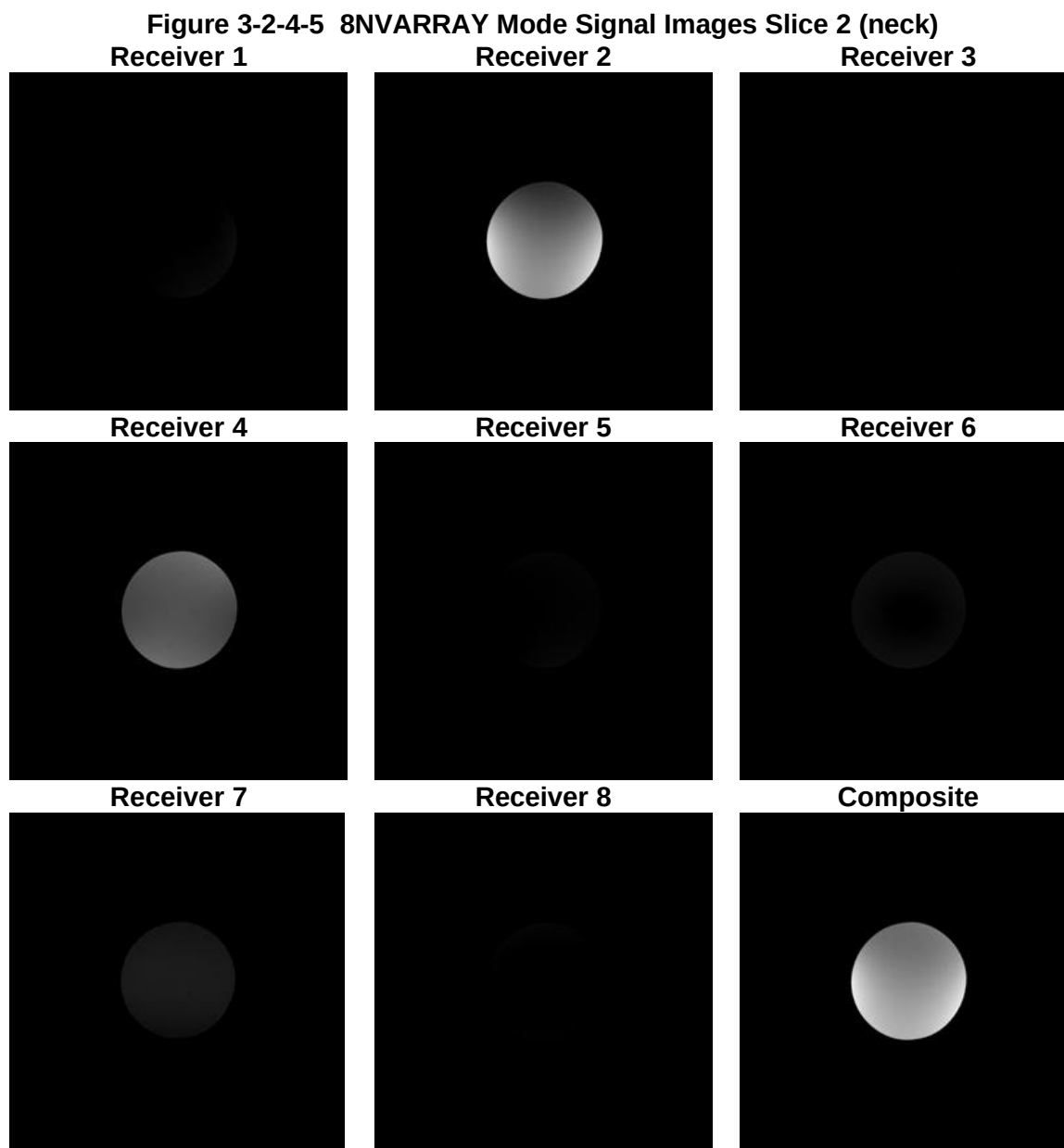
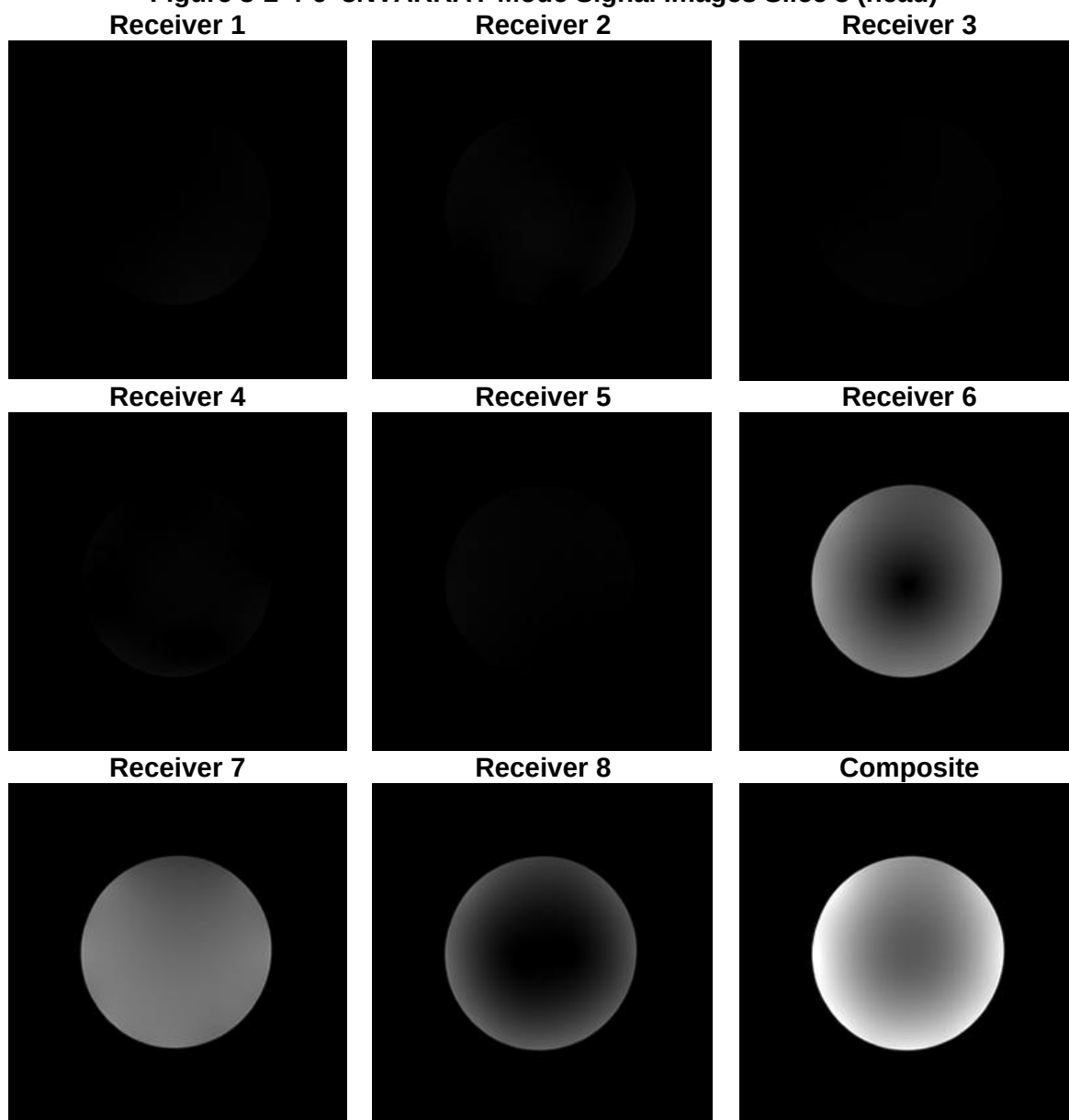


Figure 3-2-4-6 8NVARRAY Mode Signal Images Slice 3 (head)



Chapter 6 - Installation and Maintenance

Installation and Configuration

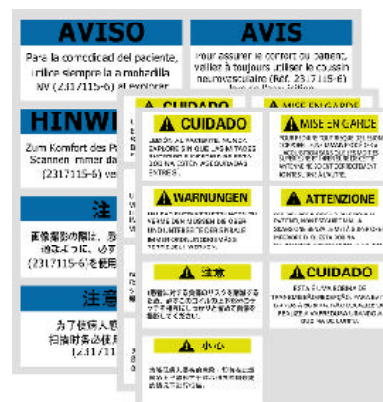
Coil installation and configuration requires the services of your GE Service Representative.

Please refer to the Service Manual for coil installation and configuration information.

Translated Safety Labels

The 8 Channel NeuroVascular Array includes a set of safety labels that have been translated into seven languages. If one of the translations would improve the safe use of the coil in your environment, use the Translated Safety Labels as follows.

- Place the appropriate blue Translated Safety Label on top of the existing blue safety label.
- Place the appropriate yellow Translated Safety Labels on top of the existing yellow safety labels.



Cleaning

The 8 Channel Neurovascular Array, and pads **must** be cleaned and stored using the following procedure:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- **Do not pour any cleaning solution directly on the coil!**
- Let the coil housing and pads dry before use.
- Store the coil in an air-conditioned scan room or equipment room.
- **Under no circumstances should the coil be placed into any type of sterilizer.**

Storage

Store the coil in an air-conditioned scan room or equipment room.

To store the coil, a storage space of 45.7 cm/18.0 in (width) x 72.4 cm/28.5 in (depth) x 42.7 cm/16.8 in (height) is required.

Replaceable Accessories

The following table lists coil accessories that may be purchased for replacement:

Description	GE Part #	MRIDC #
Wedge Pad	2317115-8	101407
HNC Pad	2317115-6	102955

For additional information, contact:

GE Medical Systems Americas
 USA 800-558-5102
 Canada 800-668-0732

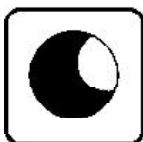
GE Medical Systems Asia
 China 86-21-62192228
 Taiwan 886-2-2505-7900
 Singapore 65-291-8528
 Australia 61-2-9975-5501
 Japan 81-120-48-2630
 Korea 82-31-740-6119
 India 91-80-845-2923

GE Medical Systems Europe
 (33) 1-41-19-76-76

GE Medical Systems Web Site
www.gemedicalsystems.com

MRI Devices, Corporation
 800-524-1476
 262-524-1402

MRI Devices Web Site
www.mridevices.com



Manufactured by:

MRI Devices Corporation
1515 Paramount Drive, Suite A
Waukesha, Wisconsin 53186 USA

Distributed by:

GE Medical Systems

